



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 • STANDARD 6.AT.1

## Understand Ratios

Lesson 3-1 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

### CHEF ACADEMY MISSION

#### **The Signature Recipe**

You just earned your apron at Chef Academy. Head Chef Reyes is trusting you with her signature cookie recipe — 3 cups of flour for every 2 cups of sugar. Before service begins, you must read every recipe like a pro by mastering ratios, or the whole kitchen falls behind.



## TODAY'S OBJECTIVES

**Content Objective:** I can write and describe a ratio that compares two quantities.

**Language Objective:** I can explain a comparison using the words ratio, part-to-part, part-to-whole, and colon notation.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Understand Ratios

Ratio

Comparison

Part-to-part

Part-to-whole

Colon notation

Tape diagram

Double number line



Tap any word to see what it means and a picture.

1

A comparison of two quantities by division is a .

2

A comparison of two quantities, like 3 to 5, is a .

3

Showing how two amounts relate to each other is a .

4

A ratio that compares one part of a group to another part is a  ratio.

5

A ratio that compares one part of a group to the total is a  ratio.

6 Writing a ratio with a colon, like 3:5, is using .

7 A bar cut into equal parts that shows how a ratio breaks apart is a

.

8 Two lines whose matching ticks pair the two amounts in a ratio form a

.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

#### EXPLAIN

**The school cafeteria is planning a new lunch menu. They want to offer 3 main dishes for every 2 side dishes. If they have 10 side dishes, the cafeteria manager needs to figure out how many main dishes to prepare?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **Understand Ratios** — Use a ratio to compare two quantities by division.  
*Comprender razones — Usar una razón para comparar dos cantidades mediante la división.*

☐ **Ratio** — A way to compare two amounts, like 3 to 2.  
*Razón — Una manera de comparar dos cantidades, como 3 a 2.*

☐ **Comparison** — Looking at two or more amounts to see how they are related.  
*Comparación — Mirar dos o más cantidades para ver cómo se relacionan.*

☐ **Part-to-part** — A ratio comparing one part of a group to another part.  
*Parte a parte — Una razón que compara una parte de un grupo con otra parte.*

☐ **Part-to-whole** — A ratio comparing one part to the whole group.  
*Parte a todo — Una razón que compara una parte con todo el grupo.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**For every \_\_\_\_ main dishes the cafeteria serves \_\_\_\_ side dishes, so with 10 side dishes they need \_\_\_\_ main dishes.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The recipe uses 3 cups of apple juice for every 2 cups of sparkling water.

**Evidence:** A strong answer names that apple juice and sparkling water are being compared and states the relationship as 3 to 2 (or 'for every 3 apple juice, 2 sparkling water'), not just that there is more apple juice.

**Reasoning:** This evidence connects the definition of understand ratios to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **For every** \_\_\_\_ **main dishes the cafeteria serves** \_\_\_\_ **side dishes, so with 10 side dishes they need** \_\_\_\_ **main dishes.**
- **My answer is** \_\_\_\_.

*Mi respuesta es* \_\_\_\_.

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is** \_\_\_\_.  
*Mi afirmación es* \_\_\_\_.
- **I know because** \_\_\_\_.  
*Lo sé porque* \_\_\_\_.
- **So** \_\_\_\_.  
*Entonces* \_\_\_\_.

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
  - **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
  - **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*
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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.



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## Answer Key & Teacher Guide

1. **Notes 1:** Understand Ratios
2. **Notes 2:** Ratio
3. **Notes 3:** Comparison
4. **Notes 4:** Part-to-part
5. **Notes 5:** Part-to-whole
6. **Notes 6:** Colon notation
7. **Notes 7:** Tape diagram
8. **Notes 8:** Double number line
9. **Watch for this mistake:** *A common mistake in Understand Ratios is writing the ratio in the wrong order: when asked for the ratio of apple juice to sparkling water in a recipe that uses 3 cups of apple juice for every 2 cups of sparkling water, some students write 2:3 (sparkling water to apple juice) because they list the ingredients in whatever order comes to mind rather than the order the question asks for. The two ratios describe different drinks — 3:2 means 3 cups of juice for every 2 cups of water, while 2:3 would mean only 2 cups of juice for every 3 cups of water. Always match the order of your ratio to the order named in the question: "apple juice to sparkling water" means the apple juice number comes first, giving 3:2.*
10. **Try It 1:** A. 5:12 — *The whole class is  $5 + 7 = 12$  students. Part-to-whole compares one part to the total, so chefs to whole class is 5:12.*
11. **Try It 2:** A. 9:4 — *"For every 9 honey to 4 lemon" is a part-to-part comparison. In colon notation, honey to lemon is 9:4.*